

SANSKAR VERMA

AI / ML Engineer • Data Engineer • Software Development Engineer

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PROFESSIONAL SUMMARY

Computer Science (AI) graduate from PSIT Kanpur with a strong foundation in Data Engineering, Machine Learning, and Backend Development. Built end-to-end systems spanning AI-powered technical interviewing, data warehousing, NLP, and REST APIs using Python, Java, SQL, Spring Boot, and Docker. Published as First Author in CRC Press (Taylor & Francis) and served as Vice President of the Aryabhata Club, leading technical events and cross-functional student teams.

PROJECTS

AI Interviewer

React, Bun, PostgreSQL, Prisma, Gemini API, Deepgram, GitHub API

[GitHub: sanskarv/AI-Interviewer-AI-Powered-Technical-Interview-Platform](#)

- Implemented an AI-powered technical interview platform (guided build) that analyzes GitHub repositories to generate personalized, project-specific interview questions using Gemini.
- Built a real-time voice interview pipeline using Deepgram speech-to-text and browser audio, persisting interview questions and candidate responses through PostgreSQL and Prisma.

Research Desk AI — Multi-Agent Research System

Python, LangChain, Groq (gpt-oss-20b), Tavily API, BeautifulSoup, Streamlit

[GitHub](#) • [Live Demo](#)

- Designed and built a 4-stage multi-agent research pipeline using LangChain — a Search agent (Tavily API) and Reader agent (web scraping) gather information, a Writer chain drafts a structured report, and a Critic chain scores and reviews it for quality.
- Implemented custom LangChain tools for web search and content scraping (BeautifulSoup), orchestrating a Groq-hosted LLM (gpt-oss-20b) across all four agent/chain stages.
- Built and deployed an interactive Streamlit interface for end-to-end topic research, from query to a critiqued, structured report.

Data Warehouse & Analytics Platform

SQL Server · ETL Pipelines · Medallion Architecture · Star Schema · Data Modelling · Notion

[GitHub: sanskarv/sql-data-warehouse-project](#)

- Designed and built a modern Data Warehouse using Medallion Architecture — Bronze layer for raw ERP/CRM data, Silver for cleansing, Gold for business-ready analytics — processing 116K+ records across two source systems.
- Built ETL pipelines to extract, transform, and load data across all three layers, resolving 50+ data quality issues (duplicate keys, corrupted records, missing values, calculation mismatches) and merging two source systems into one unified data model.
- Developed a Star Schema in the Gold layer with fact and dimension tables — 60K+ sales transactions across ~18K customers and 397 products — optimised for analytical queries, used to generate SQL reports on customer behaviour, product performance, and sales trends.
- Documented the full architecture in Notion: data flow diagrams, data models, naming conventions guide, and a data catalog for easy extensibility.

TECHNICAL SKILLS

Languages: Python, SQL, NodeJS, Java, HTML, CSS

AI / ML: NLP, Machine Learning, Sentiment Analysis, Scikit-learn, LLM, RAG(basics)

Data Engineering: ETL Pipelines, SQL Server, Star Schema, Medallion Architecture, Data Modelling, PowerBI, Snowflake(basics)

Backend & APIs: Spring Boot, REST APIs, FastAPI (basics), MySQL

Tools & DevOps: Docker, Git, GitHub, Postman, VSCode

Soft Skills: Team leadership, Event Management, Stakeholder Communication

RESEARCH PUBLICATION

IoT-Integrated Cloud and Data Solutions — CRC Press, Taylor & Francis

- Authored and published a book chapter as First Author at 21, during the third year of college - accepted by CRC Press after formal peer review, assigned a DOI (10.1201/9781003392224), and released in hardback, paperback, and e-book editions with an 89% originality score across four international editors.
- Collaborated with co-authors from PSIT Kanpur and Amity University Lucknow on topics including IoT-cloud integration, edge computing, data warehousing, data lakes, and reactive ML models - gaining early cross-institution experience on a shared technical deliverable.

EDUCATION

B.Tech — Computer Science & Engineering (Artificial Intelligence)

2021 – 2025

Pranveer Singh Institute of Technology, Kanpur

LEADERSHIP & ACTIVITIES

Vice President — Aryabhata Club, PSIT Kanpur

2021 – 2025

- Led the club as Vice President, managing a team and organising technical events including tech shows and college fest activities attended by hundreds of students.
- Coordinated logistics, speaker sessions, and cross-functional responsibilities — developed strong communication and people-management skills working with faculty and student teams.

CERTIFICATIONS & ACHIEVEMENTS

REST API Development (Udemy) · Python Programming (Springboard) · JavaScript (Springboard) · 5-Star Python Coder — HackerRank (Highest proficiency badge) · First Author Publication — CRC Press, Taylor & Francis (Peer-reviewed book chapter)